

Project Objective

The goal of this analysis was to understand **customer purchasing behaviour** for bike sales and identify the key demographic and income factors influencing purchase decisions.

Key Insights

1. Income Strongly Influences Bike Purchases

- Customers with **higher average income** are significantly more likely to purchase bikes.
- Male buyers show a slightly higher average income compared to female buyers, correlating with higher purchase rates.

2. Age Group Trends

- **Middle-aged customers (31–54)** form the largest segment of bike buyers.
- Younger (<30) and older (>55) groups show comparatively lower purchase frequency.

3. Commute Distance Impact

- Customers commuting **0–5 miles** are the most frequent bike purchasers.
- Purchase likelihood drops as commute distance increases beyond 10 miles.

4. Marital Status & Purchase Behavior

- **Married customers** account for a higher proportion of bike purchases than single customers, indicating lifestyle-driven buying decisions.

5. Gender Distribution

- While both genders purchase bikes, **male customers slightly dominate** in total purchases.
- Female buyers show consistent behavior across age and income brackets, suggesting untapped growth potential.

Actionable Recommendations

1. Target Middle-Aged Professionals

- Focus marketing campaigns on customers aged **31–54**, especially those with moderate to high income.

2. Income-Based Product Segmentation

- Offer premium models and add-ons for higher-income customers.
- Introduce value-oriented options for lower-income but high-interest segments.

3. Commute-Focused Promotions

- Promote bikes as an efficient solution for **short daily commutes (≤5 miles)**.

4. **Family & Lifestyle Messaging**

- Design campaigns around **married customers**, highlighting health, sustainability, and family benefits.

5. **Expand Female-Focused Marketing**

- Launch targeted initiatives and product designs to increase female customer engagement and conversions.